COM 3401
DATA COMMUNICATION AND NETWORKING

Lecture 08

MULTIPLEXING

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Topics covered

- Frequency-Division Multiplexing (FDM)
- Time-Division Multiplexing (TDM)
- Wavelength-Division Multiplexing (WDM)

Reference:

William Stallings, *Data and Computer Communications*, 8th Edition

Andrew S. Tanenbaum, *Computer Networks,* 5th Edition

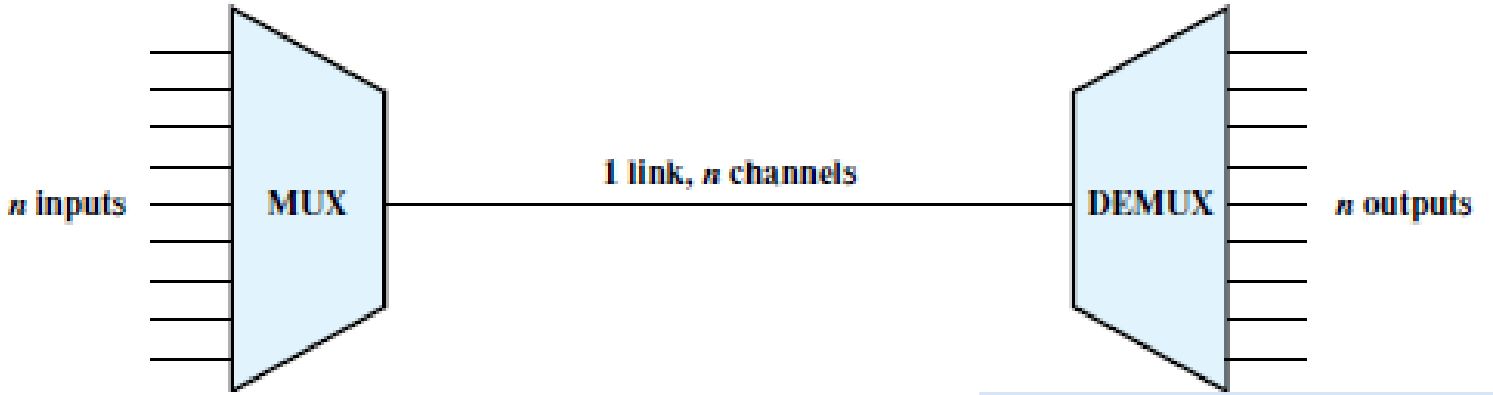
Kurose and Ross, Computer Networking – A Top-Down Approach, 6th Edition

8.1 Understanding Multiplexing



Multiplexing

- Sharing a high-capacity transmission link among multiple communication channels.



- Major types:
  - Frequency-Division Multiplexing (FDM)
  - Time-Division Multiplexing (TDM)
  - Wavelength-Division Multiplexing (WDM)
- **Basic idea:**
  - Multiple sources ➡️ Shared transmission link ➡️ Multiple destinations

Multiplexing

- The basic problem is that a transmission facility may have much more capacity than an individual communicating device requires.
- Instead of giving every source its own separate physical link, several sources can share one higher-capacity link.
- Multiplexing is a way for **several transmission sources to share a larger transmission capacity**.
  - It is particularly common in long-haul communications where high-capacity fiber, coaxial or microwave links can carry many voice and data transmissions simultaneously.

What Is Multiplexing?

- A technique that allows:
  - Several transmission sources to **share one high-capacity link**
  - Multiple channels to be carried **simultaneously**
  - A single physical link to carry **multiple logical channels**
- At the receiving end: **Demultiplexing**

Term	Meaning
MUX	Combines multiple input channels
Shared link	Carries the multiplexed signal
DEMUX	Separates the multiplexed signal

Why multiplex?

- A high-capacity transmission facility is expensive.
- But individual devices often require only a relatively small data rate.
- Therefore:
  - **Many low-rate sources → One high-rate transmission facility**
- **Benefits:**
  - Better utilization of transmission capacity
  - Lower cost per individual channel
  - Efficient use of high-capacity links
  - Supports simultaneous communication

Three Ways to Share the Link

- The transmission resource can be divided according to:

Technique	Resource divided
FDM	Frequency
TDM	Time
WDM	Wavelength

- **This is the central idea:**
  - **FDM:** users transmit at the same time but occupy different frequency bands.
  - **TDM:** users share the same frequency range but transmit during different time slots.
  - **WDM:** different optical wavelengths (colors of light) share the same optical fiber.

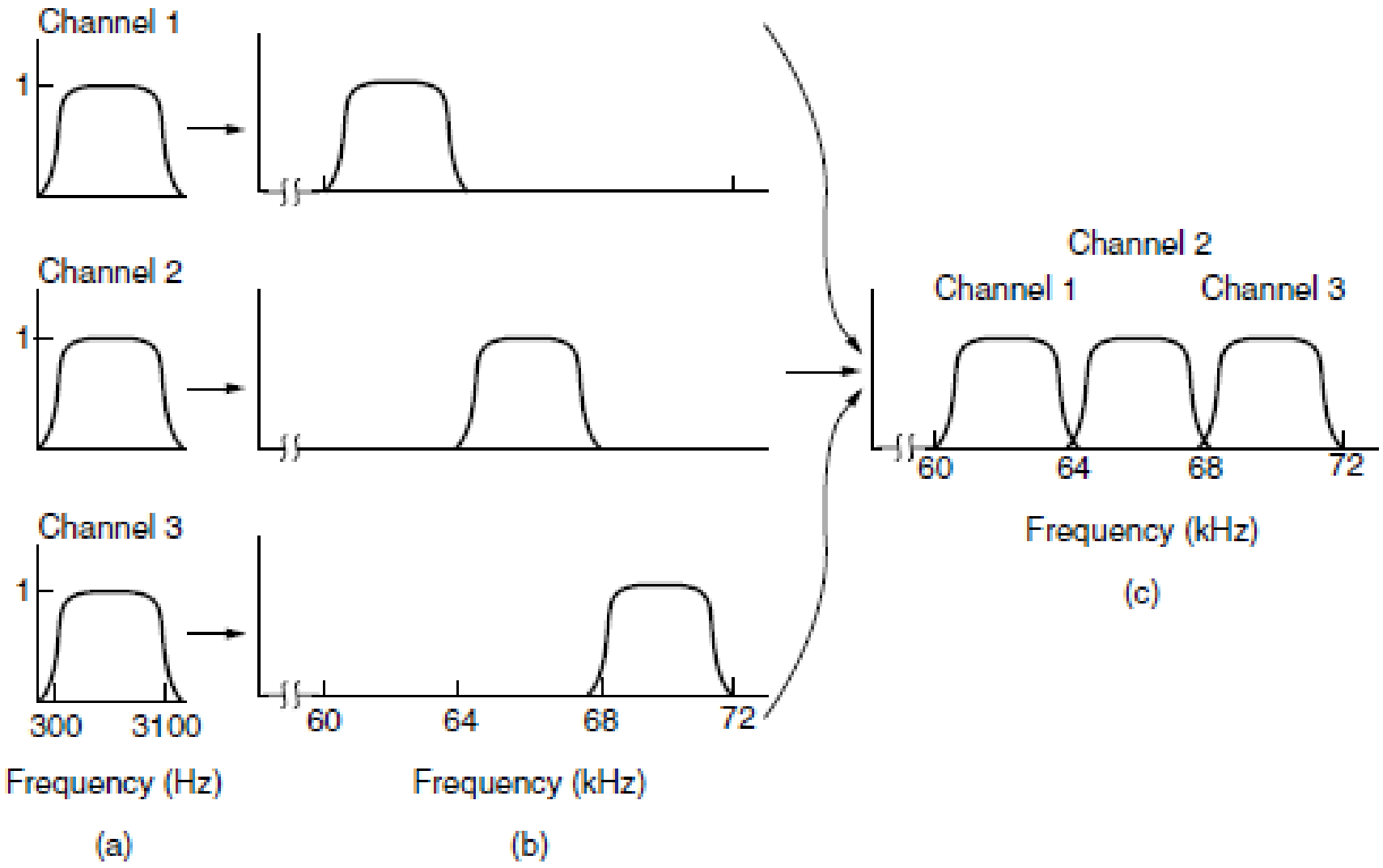
8.2 Frequency Division Multiplexing



Frequency-Division Multiplexing (FDM)

- The available frequency spectrum of a transmission medium is divided into **frequency bands**.
- Each channel is assigned a different frequency band.
- **All channels transmit simultaneously.**
- **Key idea**
  - **Different frequency → simultaneous transmission**

Frequency-Division Multiplexing (FDM)



How FDM Works

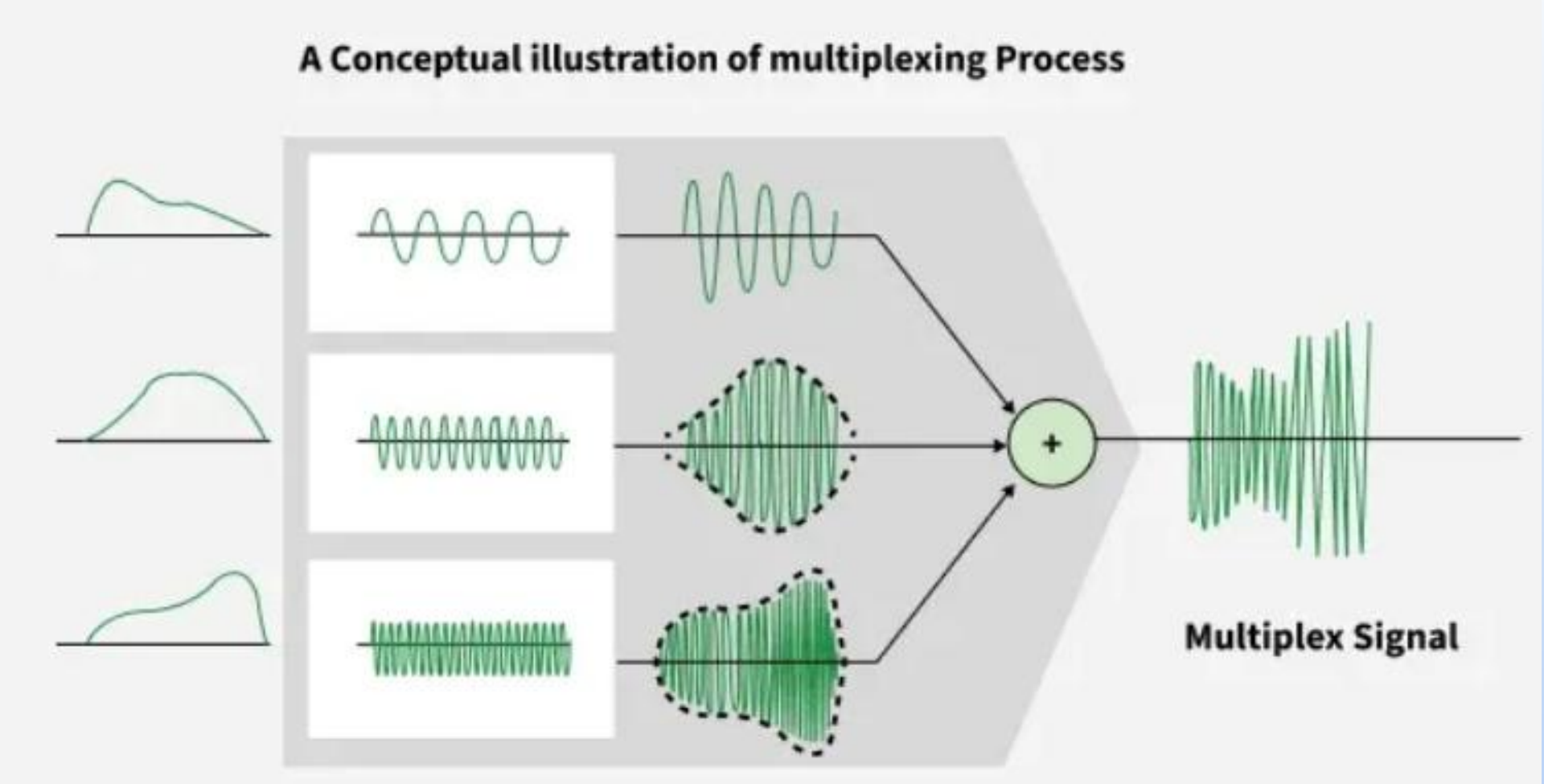
At the transmitter

- Each source produces a signal.
- Each signal is modulated onto a different carrier frequency.
- The frequency bands are positioned so they do not significantly overlap.
- The modulated signals are combined.

At the receiver

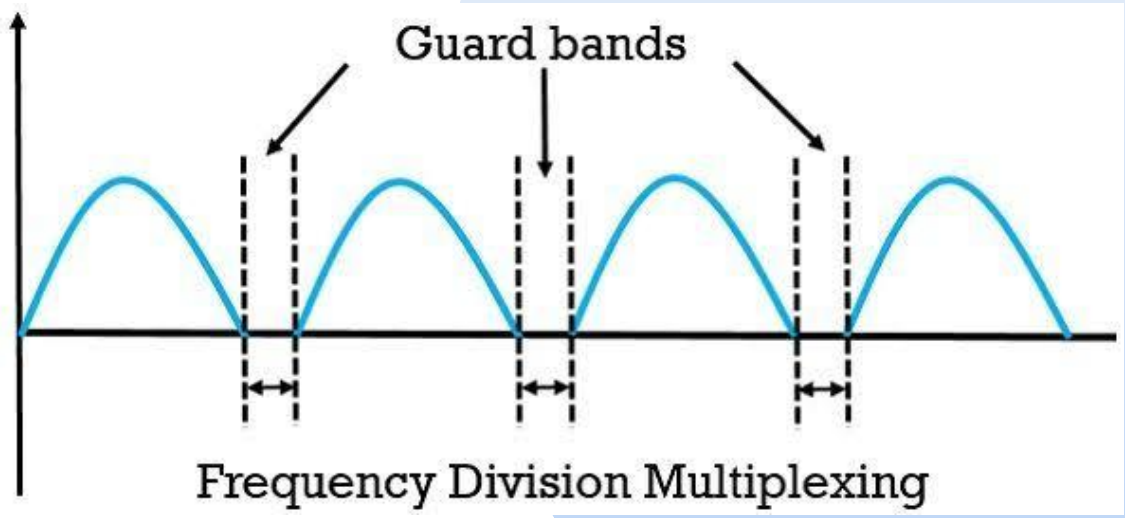
- Receive composite FDM signal.
- Use filters to select individual frequency bands.
- Demodulate each selected signal.
- Recover the original signals.

How FDM Works



Guard Bands in FDM

- Each channel occupies a portion of the available spectrum. Adjacent channels must be sufficiently separated to reduce interference.
- **Guard Band:** A small unused frequency range placed between adjacent channels.



- **Purpose**
  - Reduce interference between adjacent channels
  - Keep channel spectra sufficiently separated

FDM Example 01: Telephone Channels

Voice channel

- Allocated bandwidth ≈ **4000 Hz**
- Useful bandwidth ≈ **3100 Hz**
- Difference → **guard band**

Historically:

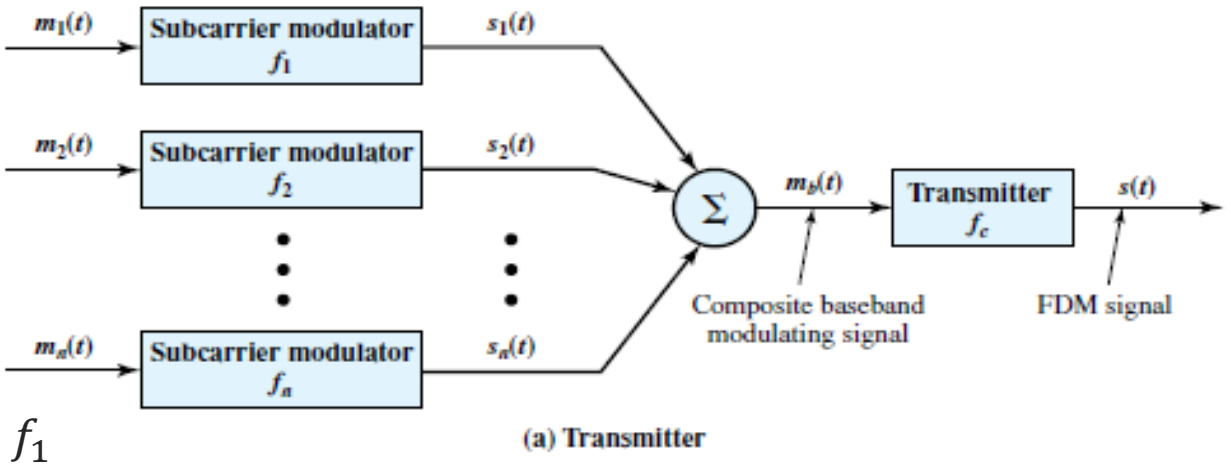
- **12 voice channels → 60–108 kHz band (12 channel group)**

Then:

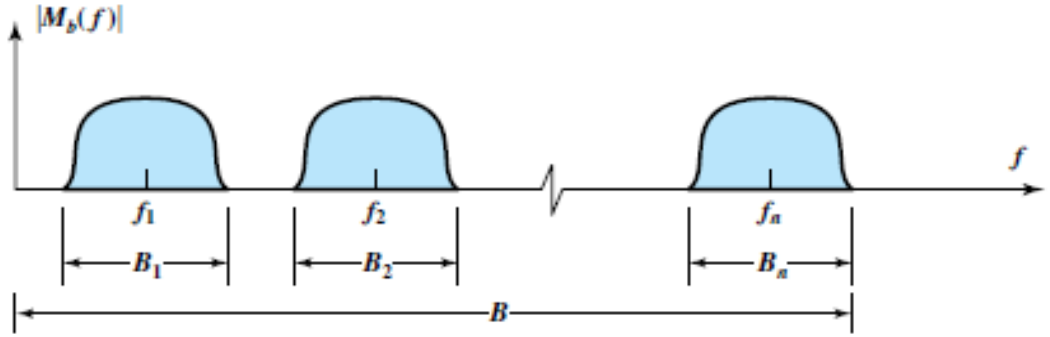
- **5 groups → 60 voice channels (60 channel supergroup)**

FDM Example 02: Three Voice Signals

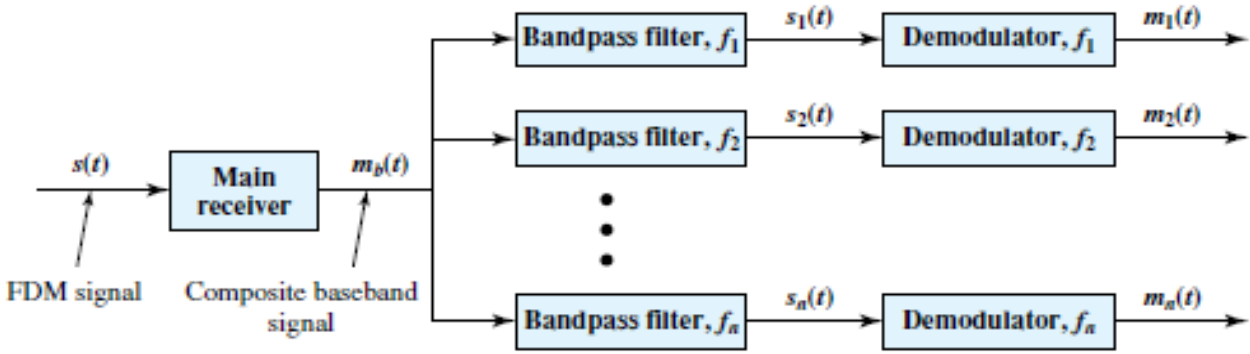
- Each voice signal:
  - **Bandwidth ≈ 4 kHz**
- Example:
  - Signal 1 ⟶ carrier frequency f_1
  - Signal 2 ⟶ carrier frequency f_2
  - Signal 3 ⟶ carrier frequency f_3
- The modulated signals are combined to form the **FDM signal**.



(a) Transmitter



(b) Spectrum of composite baseband modulating signal



(c) Receiver

Advantages and Limitations of FDM

Advantages

- Multiple channels transmit simultaneously
- Suitable for analog signals
- Each channel has dedicated frequency band
- Widely used in radio/television and telecommunications

Limitations

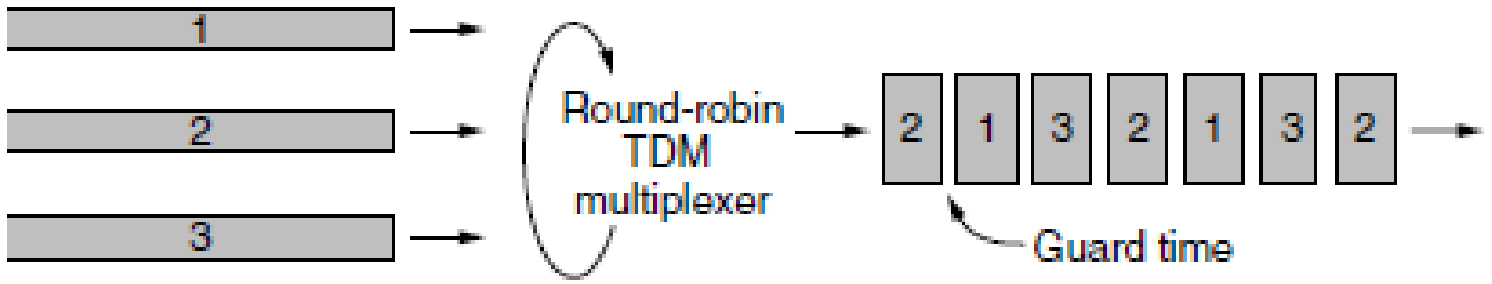
- Requires modulation/demodulation
- Guard bands consume spectrum
- Adjacent-channel interference/crosstalk can occur
- Requires filtering

8.3
Time Division Multiplexing



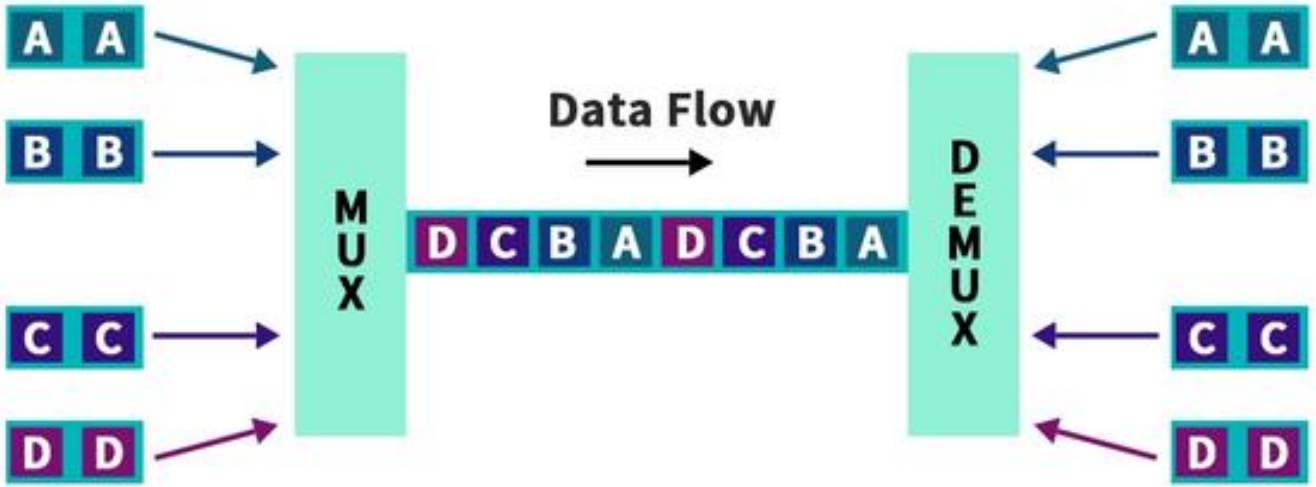
Time-Division Multiplexing (TDM)

- The transmission resource is divided into **time slots**.
- Each source is assigned a time slot.



- Unlike FDM:
  - **FDM → different frequencies**
  - **TDM → different times**
- TDM is an alternative to FDM in which users **take turns**, with each periodically receiving the entire bandwidth for a short burst of time.

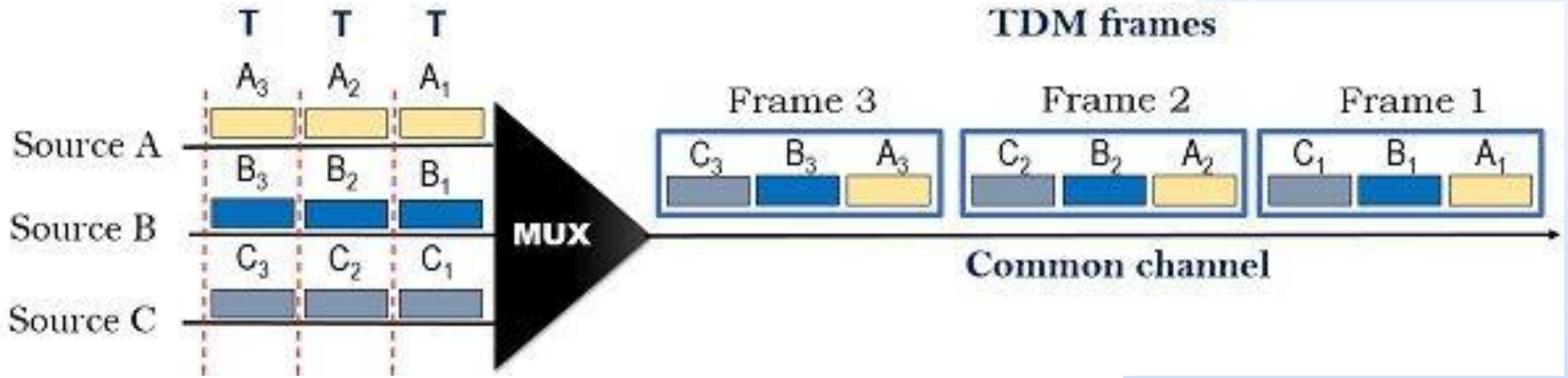
How TDM Works



- Each source receives a predetermined time slot.
- **Key idea**
  - **One channel uses the link → then another → then another → repeat**
- TDM carries multiple signals by **interleaving portions of each signal in time**. The interleaving can occur at the bit level, byte level, or larger blocks.

TDM Frames and Time Slots

- A TDM transmission is organized into **repetitive frames**.



- Each source is assigned one or more slots per frame.
- **Key idea**
  - **Same source → same position in every frame**

TDM: Data Rate of a Channel

Channel data rate in TDM channel can be calculated as follows:

Channel data rate = framerate x bits per slot

Suppose:

- Link transmits **8,000 frames/s**
- Each time slot contains **8 bits**

Then:

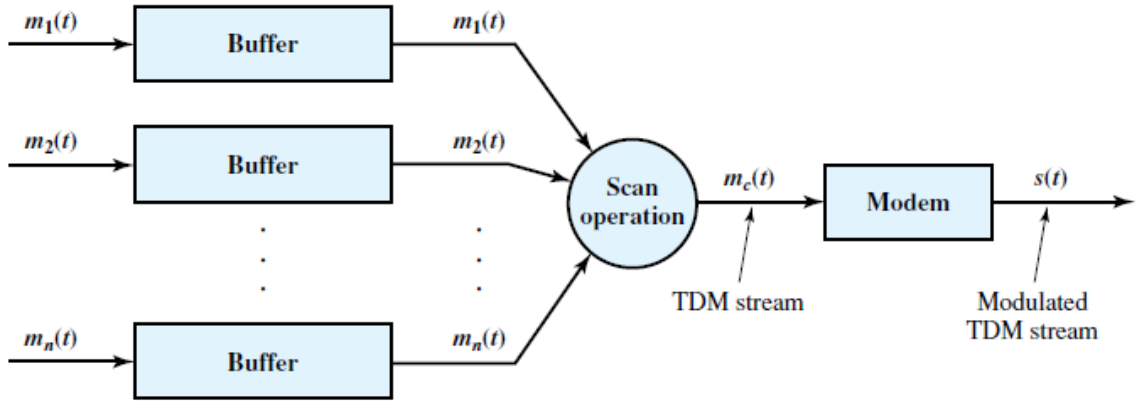
Channel data rate = 8000 x 8

=64 kbps

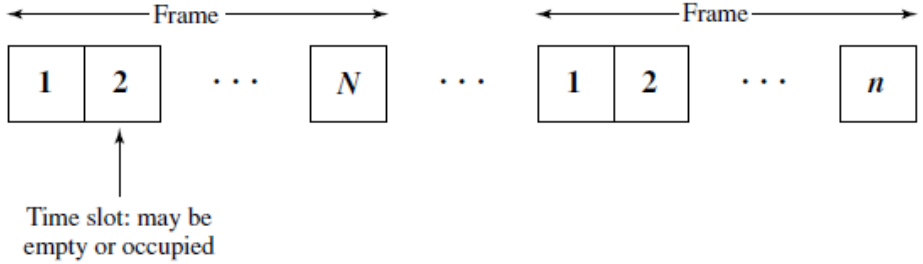
TDM Synchronization

- TDM requires the sender and receiver to remain synchronized.
- Why?
  - The receiver must know:
    - "Which bits belong to which channel?"
- **Problems if synchronization is lost**
  - Incorrect slot interpretation
  - Data from multiple channels can be affected
  - Frame boundaries may be lost

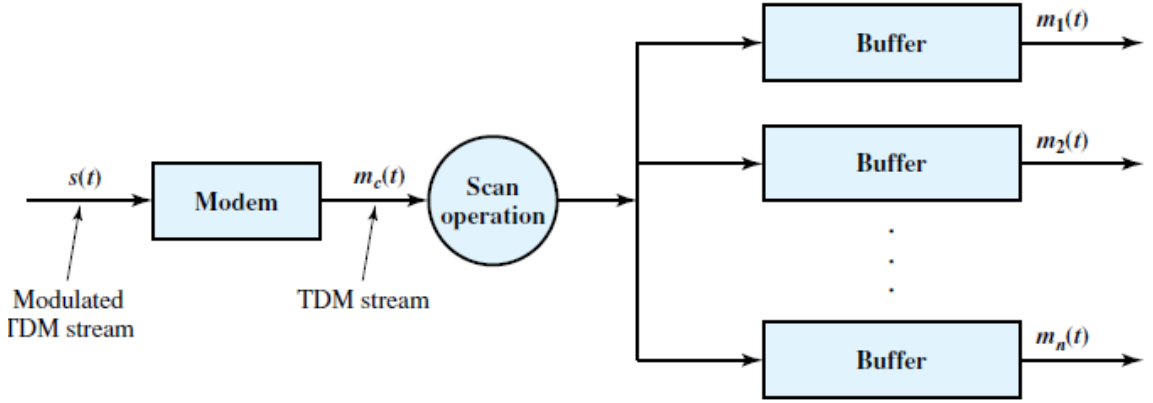
TDM Synchronization



(a) Transmitter



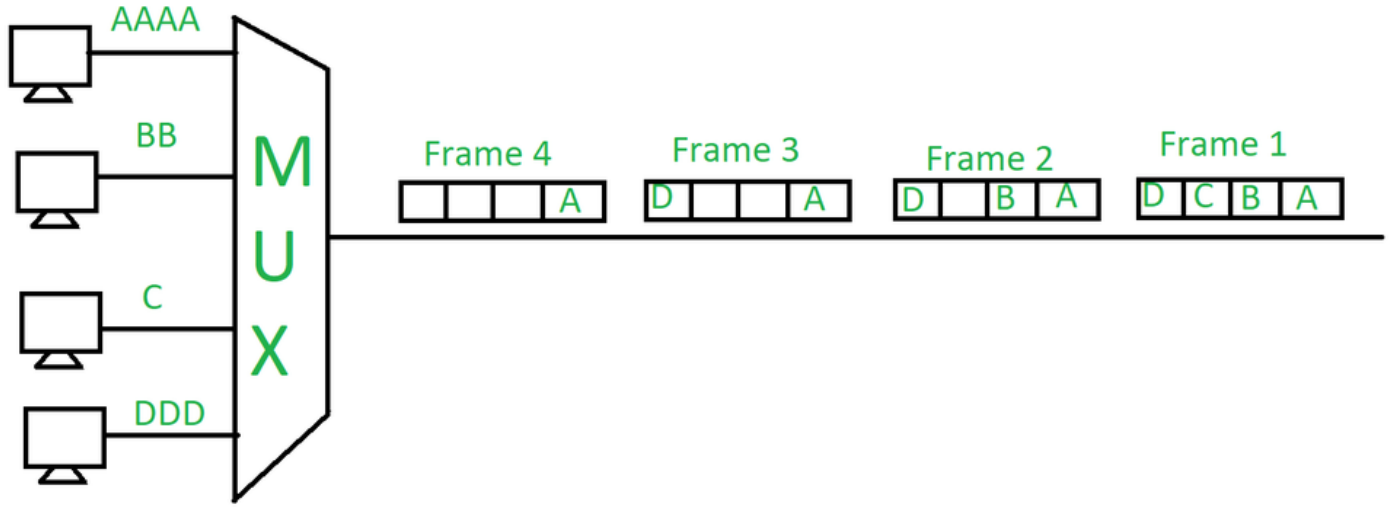
(b) TDM frames



(c) Receiver

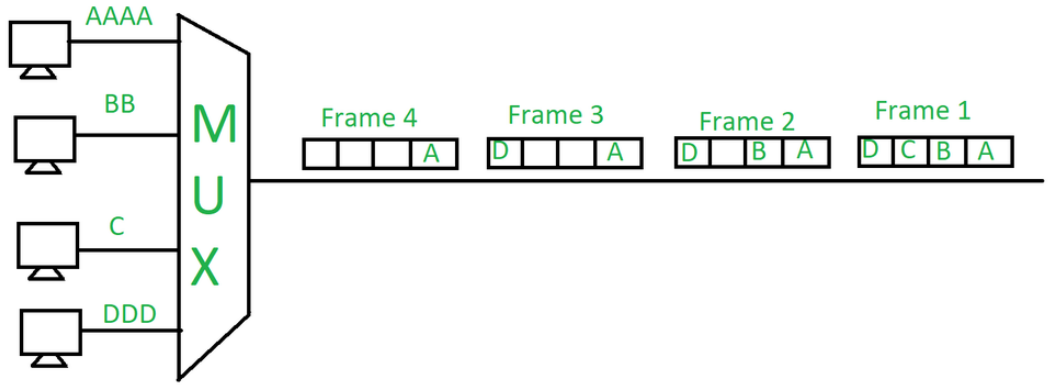
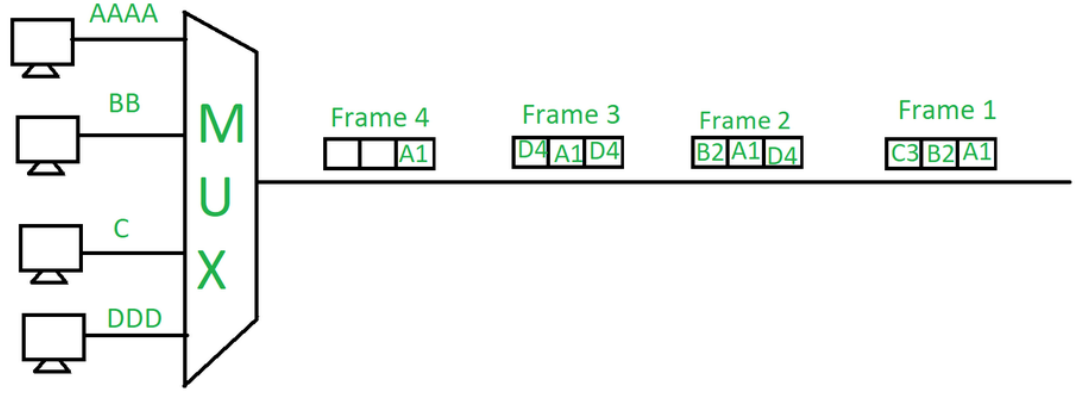
Empty Slots in Synchronous TDM

- Consider:



- The slot assigned to C may remain unused.
- **Key idea**
  - **Synchronous TDM reserves slots even when a source has nothing to transmit.**
- This leads to an **important limitation** of synchronous TDM:
  - Reserved capacity can be wasted when a source is idle.

Synchronous vs Statistical TDM

Synchronous TDM	Statistical TDM
Fixed slots	Slots allocated dynamically
Source has predetermined position	Source uses available slots
Simple	More complex
May waste unused slots	Better utilization
Predictable	Depends on traffic
	

TDM in Telephone Networks

- TDM became important because telephone networks moved toward digital transmission.
- Typical process:
  - **Analog voice → PCM → Digital samples → TDM → Transmission**
- **Example**
  - One voice channel:
    - 8,000 samples/second
    - 8 bits/sample
- Therefore:

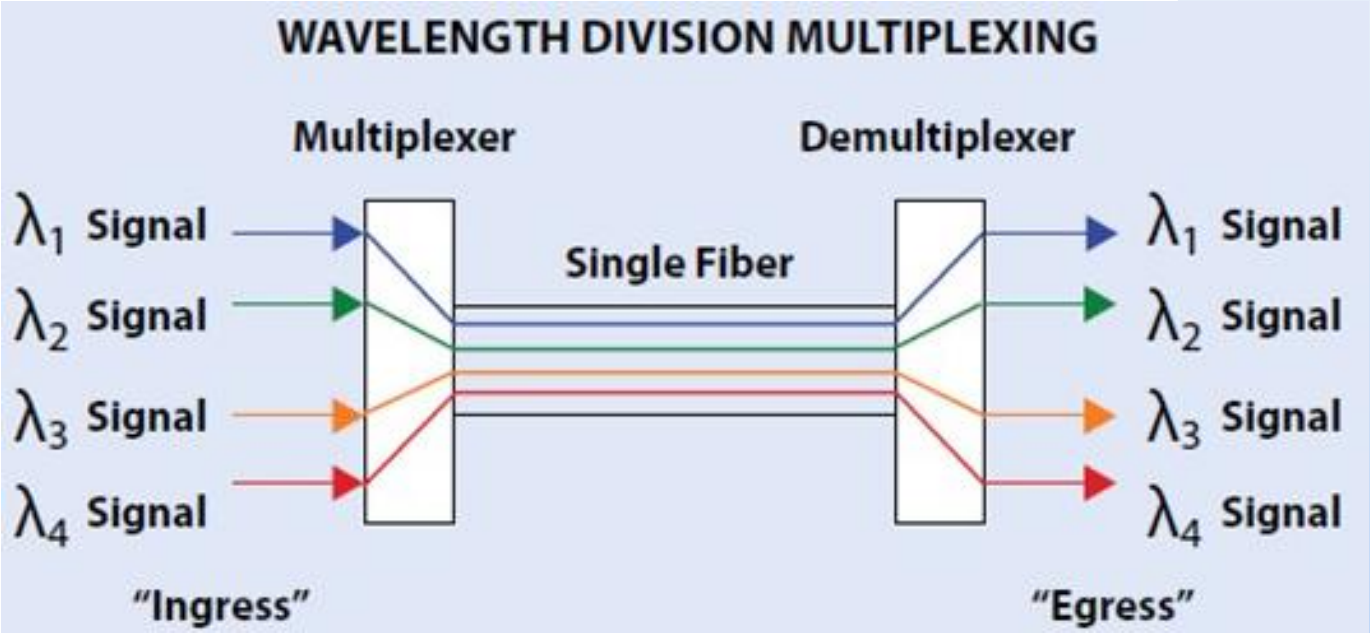
$$8000 \times 8 = 64\text{kbps}$$

8.4 Wavelength Division Multiplexing



Wavelength-Division Multiplexing (WDM)

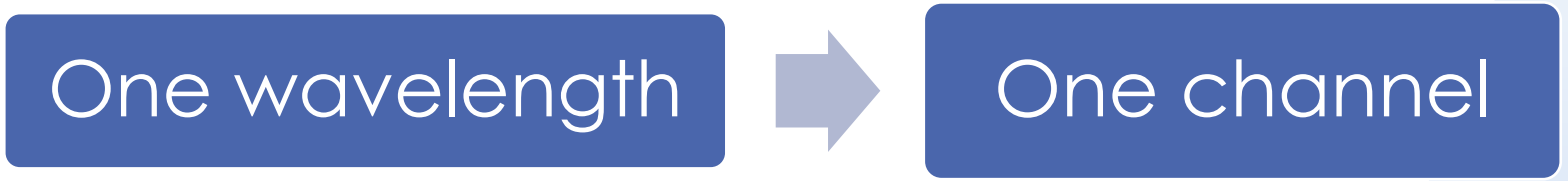
- A multiplexing technique used with **optical fiber**.
- Multiple optical signals with different wavelengths are transmitted simultaneously over **one fiber**.



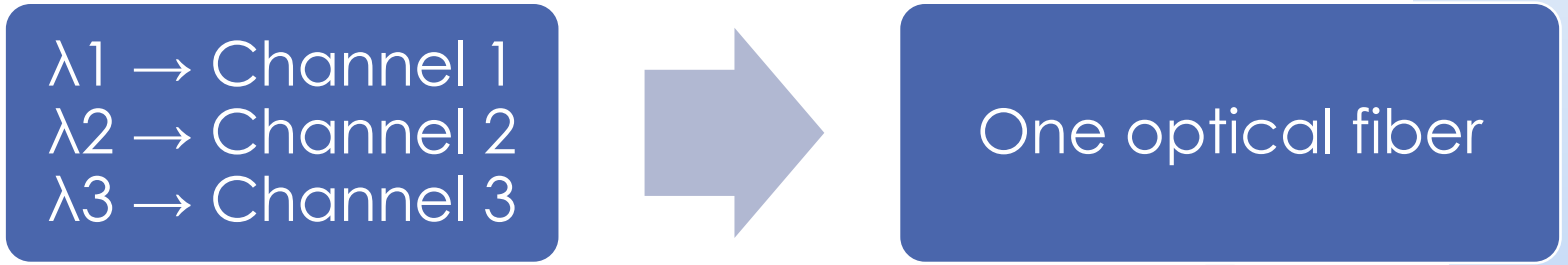
- **Key idea**
  - **Different wavelengths = different channels**

WDM: Why Use Different Wavelengths?

- A single optical fiber has enormous available bandwidth.
- Instead of using one optical signal:



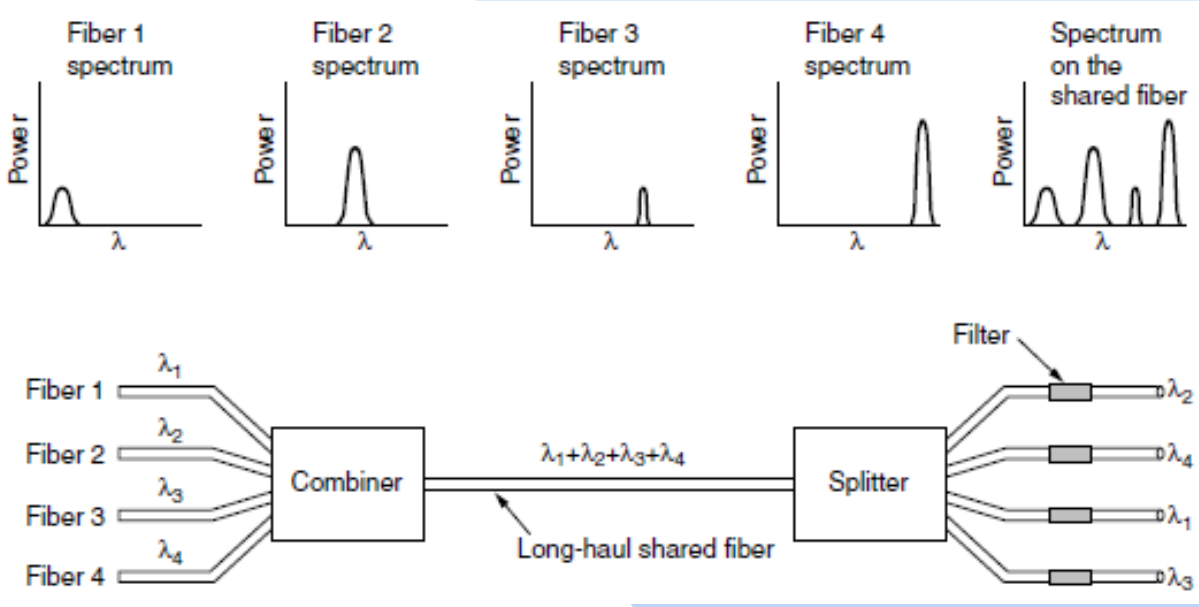
- WDM allows:



- **Key idea**
  - **More wavelengths → more parallel channels → higher aggregate capacity**

WDM Operation

- **Transmitter**
  - Multiple sources produce laser beams at different wavelengths.
- **Multiplexer/Combiner**
  - Combines the wavelengths into one optical signal.
- **Fiber**
  - All wavelengths travel together.
- **Demultiplexer/Splitter**
  - Separates the wavelengths.
- **Receiver**
  - Each wavelength is directed to its corresponding receiver.



WDM Is a Form of FDM

Relationship

- **FDM**
  - Different frequencies ➡️ Multiple channels
- **WDM**
  - Different optical wavelengths ➡️ Multiple optical channels

- **Important point**
  - WDM is essentially:
    - **Frequency-division multiplexing at optical frequencies**

WDM and Dense WDM

WDM

- Multiple wavelengths share one fiber.

DWDM — Dense WDM

- More wavelengths
- Smaller spacing between channels
- Higher aggregate capacity

WDM Capacity

- Demonstrating the capacity of WDM, assume:
  - No of beams = 100
  - Capacity of each beam = 10Gbps
  - Capacity of WDM:

$$100 \text{ beams} \times 10 \text{ Gbps}$$

$$= \mathbf{1\ Tbps}$$

- The important concept is:
- **WDM increases aggregate capacity by carrying multiple optical channels simultaneously over the same fiber.**

8.6 Comparison



FDM vs TDM vs WDM

Feature	FDM	TDM	WDM
Resource divided	Frequency	Time	Wavelength
Transmission	Simultaneous	Turn-taking	Simultaneous
Typical medium	Copper/wireless	Digital links	Optical fiber
Channel separation	Frequency bands	Time slots	Optical wavelengths
Main requirement	Frequency separation	Synchronization	Optical wavelength separation
Example	Radio/telephone	Digital telephone	Fiber-optic backbone

One Concept, Three Ways

- **FDM**
  - Everyone speaks simultaneously, but each uses a different frequency.
- **TDM**
  - Everyone uses the same resource, but takes turns.
- **WDM**
  - Multiple optical “colors” travel simultaneously through the same fiber.

Multiplexing in Real Networks

FDM

- Radio
- Television
- Telephone networks

TDM

- Digital telephone networks
- Digitized voice
- Digital transmission systems

WDM

- Long-haul optical fiber
- High-capacity optical networks

Review Questions

Conceptual

1. What is multiplexing?
2. Why is multiplexing economically useful?
3. What are the functions of a MUX and DEMUX?
4. What is the fundamental difference between FDM and TDM?
5. How does WDM relate to FDM?

FDM

6. How does FDM divide a transmission medium?
7. Why are guard bands required?
8. What is the purpose of modulation in FDM?
9. How does the receiver separate FDM channels?
10. Give one practical application of FDM.

Review Questions

TDM

11. What is a time slot?
12. What is a TDM frame?
13. Why must TDM systems be synchronized?
14. What happens when a source has no data in synchronous TDM?
15. A TDM system sends 8,000 frames/s and gives a channel 8 bits per frame. What is the channel data rate?

WDM

16. What is WDM?
17. Why is WDM particularly suitable for optical fiber?
18. What is the role of the optical combiner?
19. What happens at the demultiplexer?
20. What is DWDM?

Review Questions

Application / Thinking

21. If several channels must transmit simultaneously, which multiplexing approach is appropriate?
22. If different users take turns using a transmission resource, which technique is being used?
23. Why can WDM dramatically increase the capacity of a single optical fiber?
24. Compare FDM, TDM and WDM in terms of the resource being divided.
25. Explain multiplexing using the MUX → shared link → DEMUX model.

Thank you

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